

WASP-50b

R-1081

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_151230 · created 2026-07-30T15:32:18+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457386.644 +/- 0.052 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.171 +/- 0.05
Transit depth (Rp/Rs)^2	2.94 +/- 1.73 [%]
Orbital Inclination (inc)	86.05 +/- 2.85
Airmass coefficient 1 (a1)	13.11 +/- 7.83
Airmass coefficient 2 (a2)	-1.88 +/- 0.79
Scatter in the residuals of the lightcurve fit is	50.47 %
Best Comparison Star	#1 - [378, 332]
Optimal Aperture	3.0
Optimal Annulus	6.5
Transit Duration (day)	0.06 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.171 ± 0.05 vs 0.1404 (deviation 0.47σ)
Duration fit vs published	0.06 d vs 0.0754 d (deviation 0.41σ)
Mid-time O–C	28.0 min
Depth SNR	2.6
Residual scatter	50.47 %

Light curve

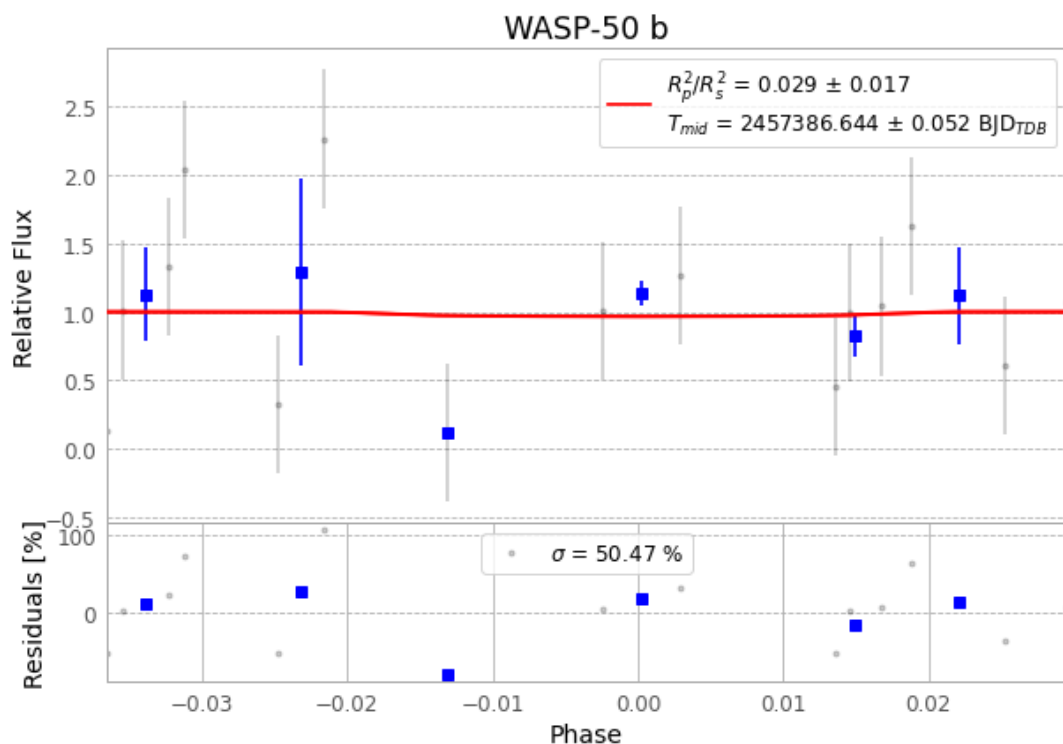


Figure 1 — FinalLightCurve_WASP-50 b_2015-12-29.png (SHA-256 978881f59aa14212...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [490, 192], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 416490b074cdf970...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

67 FITS frames (44 MB), WASP-50151230013912.FITS → WASP-50151230045712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-151230.log (d7fa62a21aed...), FinalLightCurve_WASP-50 b_2015-12-29.png (978881f59aa1...), FinalLightCurve_WASP-50 b_2015-12-29.csv (2ee196a5f72c...)

Compute resources

Wall time 190.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 15:32Z.